11.8 Tilted Equilibria

WORKSHEET

The question

STEP III 2025, Question 9

Read the question before you go on. It is in the STEP III paper in OCR's download of the 2025 STEP papers (a zip file):

www.ocr.org.uk/Images/750486-2025-past-papers.zip

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself A solid rests on a table, touching at a single point X of its curved surface. The forces on it are its weight (acting at the centre of mass G) and the reaction at X. When are these in equilibrium?

STAGE 1 The centre of mass

Covers: Part (i)

Ask yourself Slice the solid into thin cylindrical shells of radius x, thickness δx and height y. What is the volume of a shell, and how high is its centre of mass?

Ask yourself Integrate to find the volume and the moment about the base, then divide.

STAGE 2 Where normals meet the axis

Covers: Part (ii)

Ask yourself Find the gradient of the curve at $x = rp$, and so the gradient of the normal. Follow the normal to the y-axis: by how much does y change?

Pause and try For $n = 4$, differentiate your Y with respect to p to find its greatest value. What happens as $p \to 0$ and at $p = 1$?

STAGE 3 The spinning top

Covers: Part (iii), the centre of mass

Ask yourself The solid is made of two pieces from Part (i), one of them upside down. Use your volumes and centres of mass to find G.

STAGE 4 Counting equilibria

Covers: Part (iii), the tilted equilibria

Pause and try The solid can rest touching at a point exactly when the normal there passes through G. On each piece of the surface, how many points have normals that meet the axis at the height of G? Use your answer to Part (ii).

What's really going on

Equilibria as critical points

When the solid rests on the point X, the height of G above the table is the distance from G to the tangent plane at X, which for a convex body is $|GX|$ when GX is along the normal. In general, the equilibria are the *critical points* of the distance from G to the surface: a point X where $|GX|$ is stationary is exactly a point whose normal passes through G. Stable equilibria are local minima of this distance (G as low as possible), and unstable ones are local maxima or saddle points.

Going round the cross-section from A to B, the distance from G is: large at A (a maximum: balanced on its point, unstable), smaller at the tilted point on the base (a minimum, stable), larger at the rim, smaller at the lower tilted point of the dome (stable), larger at the upper one (unstable), and smaller at B (stable: resting on the top of the dome). Minima and maxima must alternate, as the figure shows. (In fact there is one more equilibrium that the question does not count: the solid can balance, unstably, on its sharp rim, where the surface has no single normal.)

The Gömböc

How few equilibria can a convex body have? Any body has at least one stable and one unstable position: the nearest and furthest points of the surface from G. A two-dimensional convex shape of uniform density always has at least four (two minima and two maxima of distance, alternating). In 1995 Vladimir Arnold asked whether a three-dimensional convex body of uniform density could have just two: one stable, one unstable. In 2006 Gábor Domokos and Péter Várkonyi constructed one, the *Gömböc*: put it down in any position, and it rolls and rocks until it reaches its single stable position. Domokos later noticed that some tortoises have shells close to this shape, which helps

them right themselves when turned over.

Archimedes and floating bodies

The study of the stability of bodies goes back to Archimedes, who in *On Floating Bodies* analysed when a floating paraboloid (a model of a ship's hull) is stable upright and when it capsizes, in terms of its shape and density. His answer, worked out with geometry alone, involves exactly the kind of centre-of-mass and normal calculations used here. Naval architects still use his ideas, in the form of the *metacentre*: a ship is stable if the point where the vertical through the centre of buoyancy meets the ship's axis lies above the centre of mass.

Try this next

1. **A self-righting toy.** Try STEP II 1996, Question 9 (step.maths.org/questions/96-s2-q9): when does a cone on a hemisphere right itself? (*Hint:* where must G be relative to the centre of the hemisphere?)

2. **A sphere.** For a uniform solid sphere, every normal passes through the centre. What does that say about its equilibria? What if the sphere's centre of mass is displaced slightly from its centre?

3. **An ellipse.** Show that an elliptical disc rolling in its own plane on a line has exactly four equilibria, and identify which are stable.

4. **Two paraboloids.** Replace the dome in part (iii) by a paraboloid $ry = r^2 - x^2$, so that the solid is symmetric about the x-axis. Show that G is at the origin, find the tilted equilibria, and explain why they come in pairs with equal tilts.

Further reading

- **Péter Várkonyi & Gábor Domokos, "Static equilibria of rigid bodies: dice, pebbles, and the Poincaré–Hopf theorem"**, *Journal of Nonlinear Science* 16 (2006). The Gömböc paper.
- **Daniel Kleppner & Robert Kolenkow, *An Introduction to Mechanics*** (Cambridge University Press), on equilibrium and stability.
- **Chris Rorres, "Completing Book II of Archimedes's *On Floating Bodies*"**, *Mathematical Intelligencer* 26 (2004). Archimedes' stability analysis, with modern pictures.

The full guided solution is in Section 11.8:
https://beyondstep.keithmcnulty.org/ch11/tilted-equilibria.html